Exercise 1: Configure the Domain Controller

Rename the Server:

- Open PowerShell as Administrator.
- Rename the server to WinServer-DC.
- Restart the server.

powershell code

- Rename-Computer -NewName "WinServer-DC" -Restart
- Go to Network and Sharing Center.
- Set a static IP address (e.g., 192.168.0.100), subnet mask (255.255.255.0), and DNS server (127.0.0.1).

Install Active Directory Domain Services (AD DS):

- Open Server Manager.
- Click 'Add roles and features'.
- Select 'Active Directory Domain Services'.
- Complete the wizard and install AD DS.

Promote to Domain Controller:

- After the AD DS role is installed, click the notification to promote the server.
- Choose 'Add a new forest' and use LabDomain.local as the root domain.
- Create a DSRM password and complete the wizard.
- Restart the server once the domain controller promotion is complete.

Install DNS Server Role:

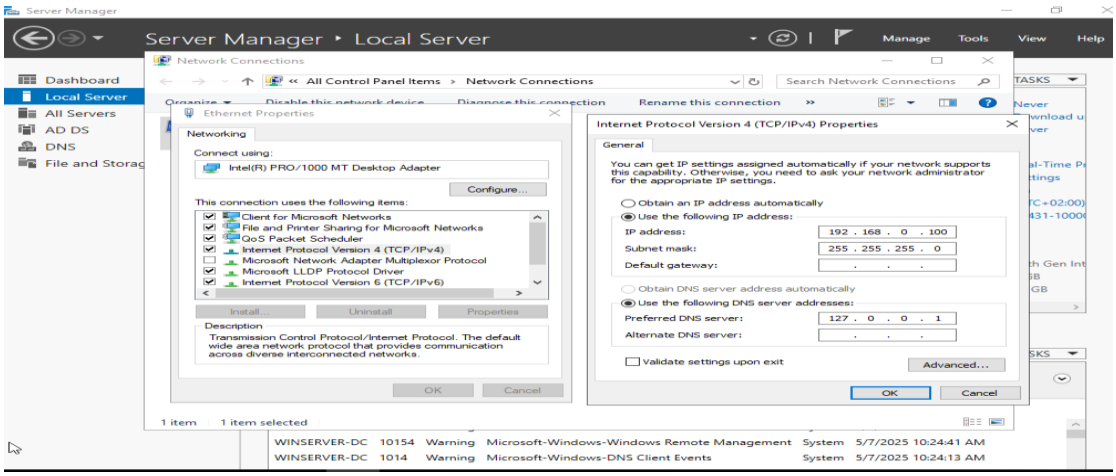
- Open Server Manager.
- Add the DNS Server role if not already installed during AD DS installation.

Evidence Required:

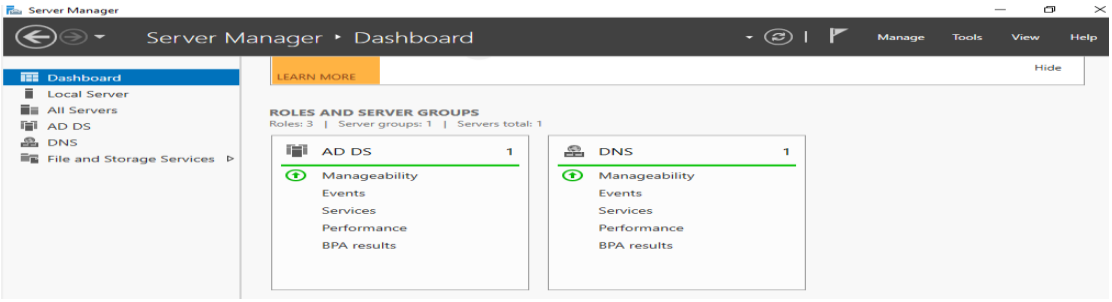
- Screenshot of WinServer-DC with the static IP configuration.
- Screenshot of Server Manager showing AD DS and DNS roles installed.
- Screenshot of Active Directory Users and Computers with the domain LabDomain.local.

Exercise 1: Configure the Domain Controller

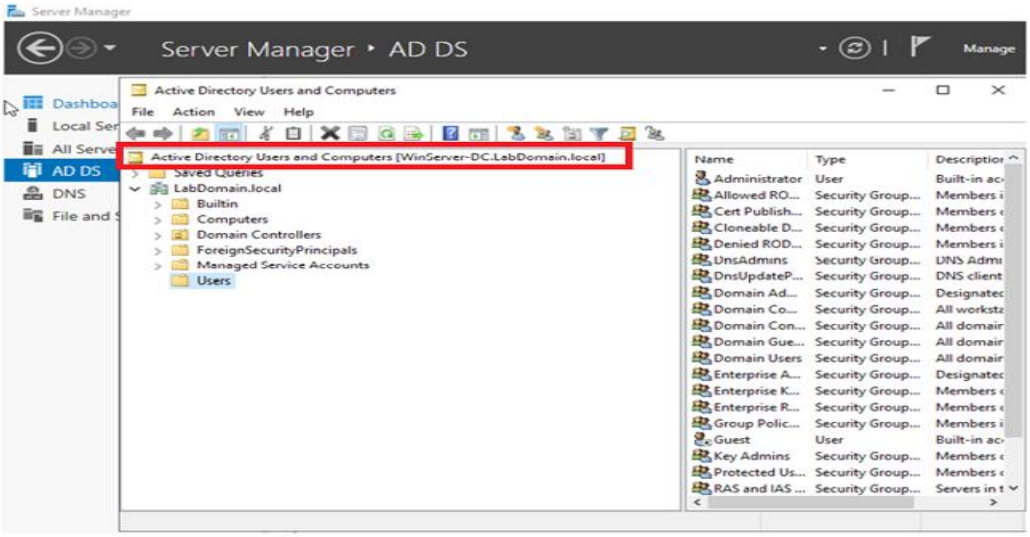
WinServer-DC with the static IP configuration
IP - 192.168.0.100 , Subnet Mask - 255.255.255.0 , DNS Server - 127.0.0.1



Server Manager showing AD DS and DNS roles installed.



Active Directory Users and Computers with the domain LabDomain.local.



Exercise 2: Configure the Client and Join the Domain

Set Static IP Address on Client:

- Configure the client to use a static IP address (e.g., 192.168.0.20), subnet mask (255.255.255.0), gateway (192.168.0.1), and DNS (192.168.0.1).

Rename the Client:

- Rename the computer to 'WinClient-01' and Restart the client.

powershell code:

- Rename-Computer -NewName "WinClient-01" -Restart

Join the Domain:

- Open 'System Properties' and change the computer name.
- Click 'Change' and join the computer to the LabDomain.local domain.
- Restart the computer and log in with domain credentials.

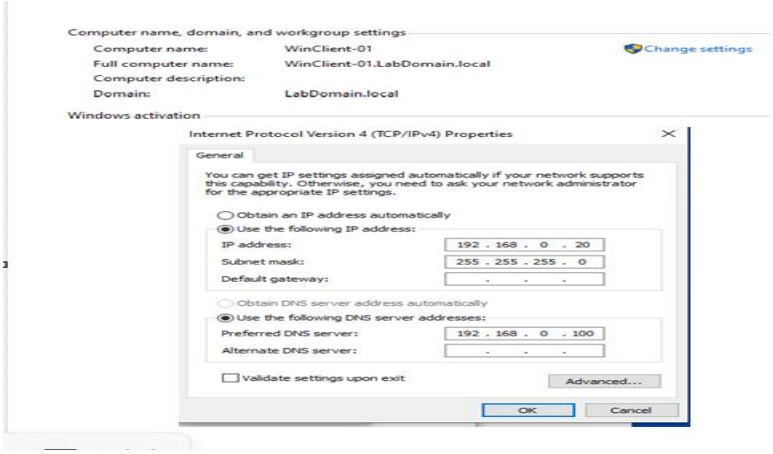
Evidence Required:

- Screenshot of WinClient-01 with the static IP configuration.
- Screenshot of 'System Properties' showing the computer joined to LabDomain.local.

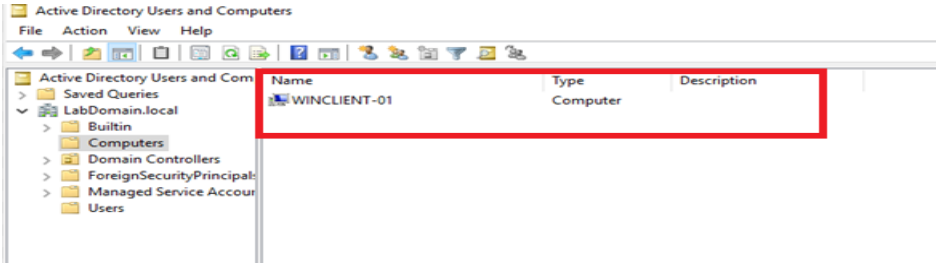
Exercise 2: Configure the Client and Join the Domain

WinClient-01 with the static IP configuration and System Properties.

IP - 192.168.0.20 , Subnet Mask - 255.255.255.0 , DNS Server - 192.168.0.100



The computer showing on windows server



Exercise 3: Create Organizational Units and Users

Create Organizational Units (OUs):

Open 'Active Directory Users and Computers'.

Create the following OUs:

- Sales
- IT
- HR

Create User Accounts:

Sales OU:

- Create two users: User1 and User2.

IT OU:

- Create one user: ITAdmin.

HR OU:

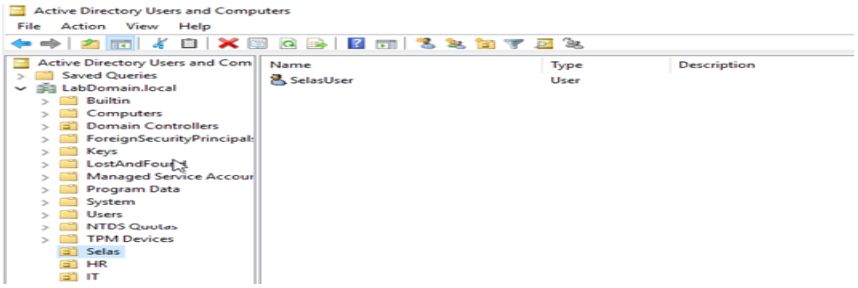
- Create one user: HRUser.

Evidence Required:

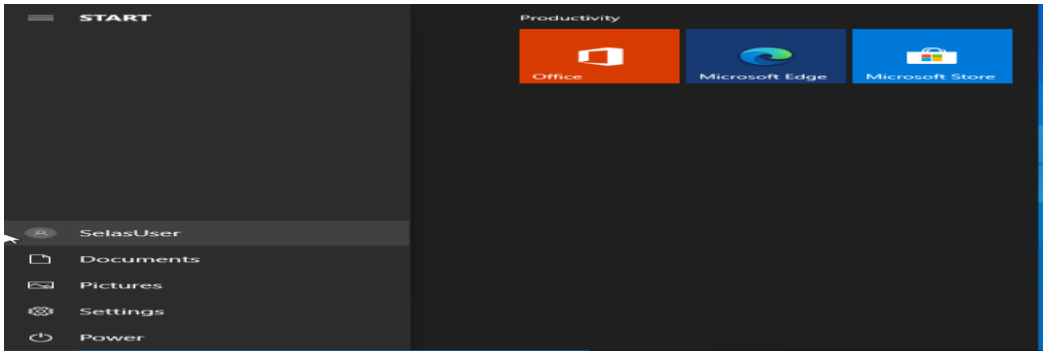
- Screenshot of Active Directory Users and Computers showing the OUs and user accounts.

Exercise 3: Create Organizational Units and Users

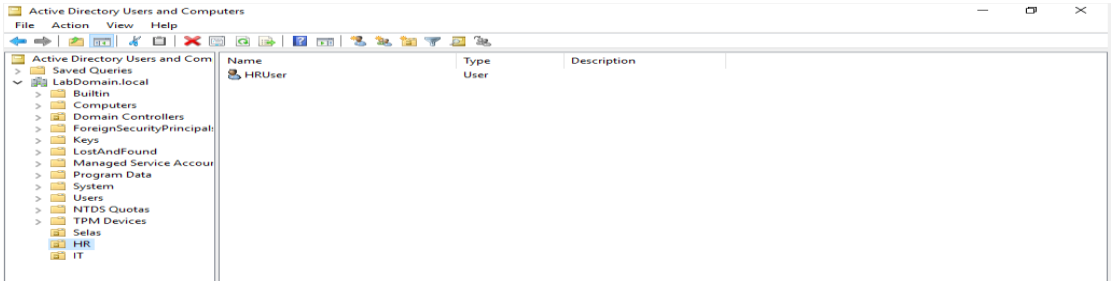
SelasUser on Active Directory Users and Computers



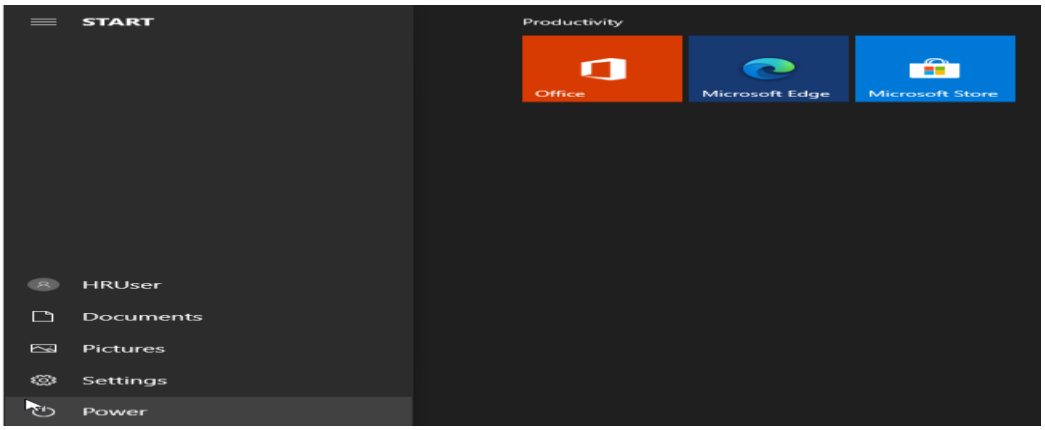
SelasUser on User Account



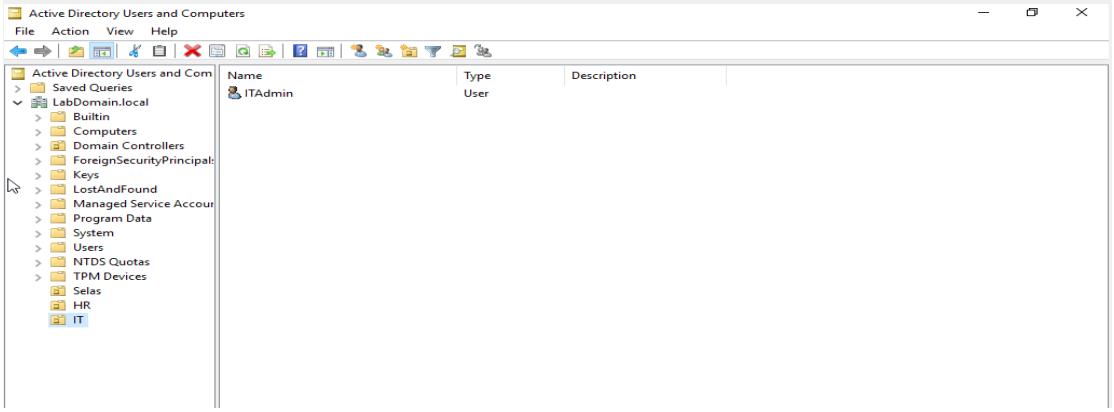
HRUser on Active Directory Users and Computers



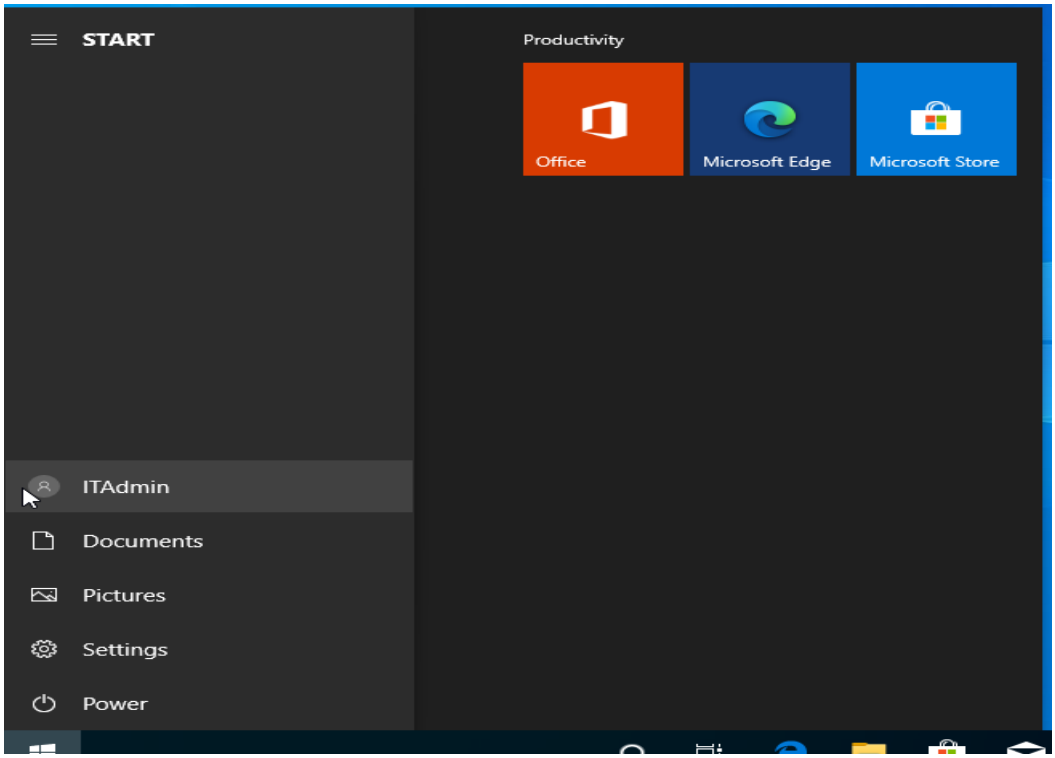
HRUser on User Account



ITAdmin on Active Directory Users and Computers



ITAdmin on User Account



Exercise 4: Configure Group Policies

Create and Apply a Password Policy GPO:

- Open 'Group Policy Management Console (GPMC)'.
- Edit the 'Default Domain Policy':
- Navigate to Computer Configuration -> Policies -> Windows Settings -> Security Settings -> Account Policies -> Password Policy.
- Set Minimum Password Length to 12, Password Complexity Requirements to Enabled, and Maximum Password Age to 60 days.

Create a Logon Script GPO:

- In GPMC, create a new GPO named Logon Script.
- Edit the GPO and add a logon script:
- Navigate to User Configuration -> Policies -> Windows Settings -> Scripts (Logon/Logoff).
- Add a logon script to start Calculator (logon_script.ps1):

File provided named: 'test.ps1'

Link the GPOs:

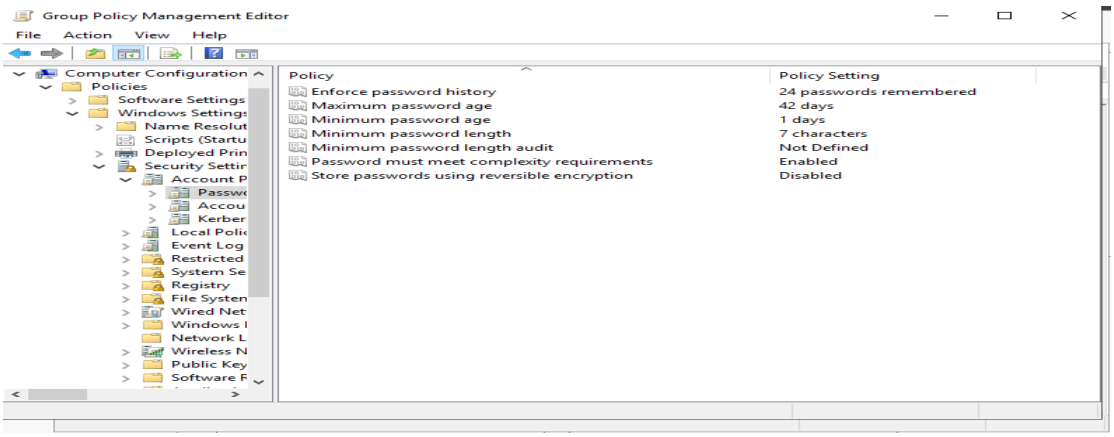
- Link the Logon Script GPO to the domain LabDomain.local.

Evidence Required:

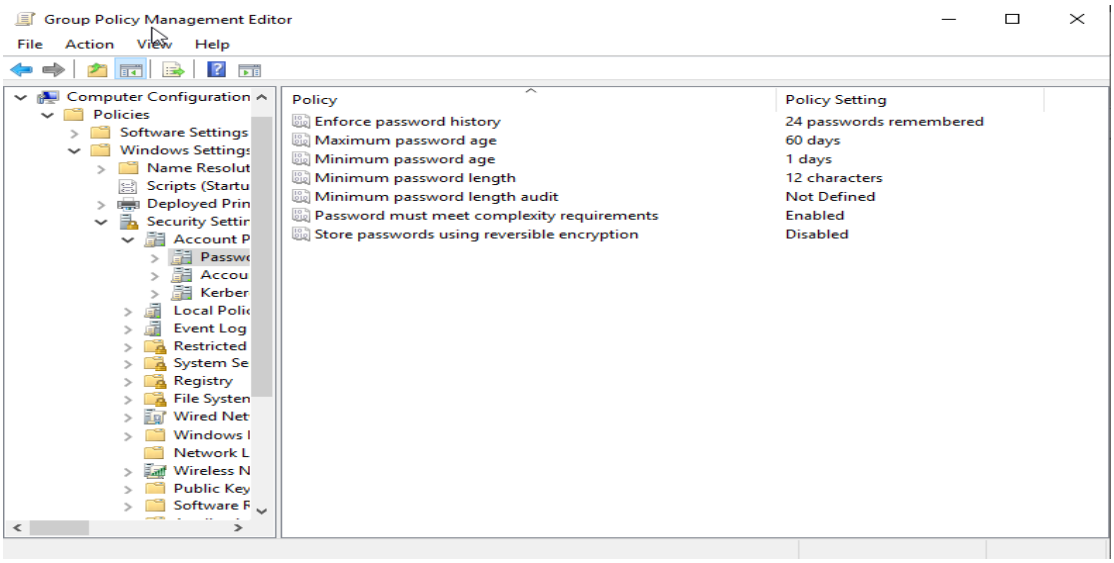
- Screenshot of GPMC showing the Default Domain Policy with the configured password policy.
- Screenshot of Logon Script GPO settings.

Exercise 4: Configure Group Policies

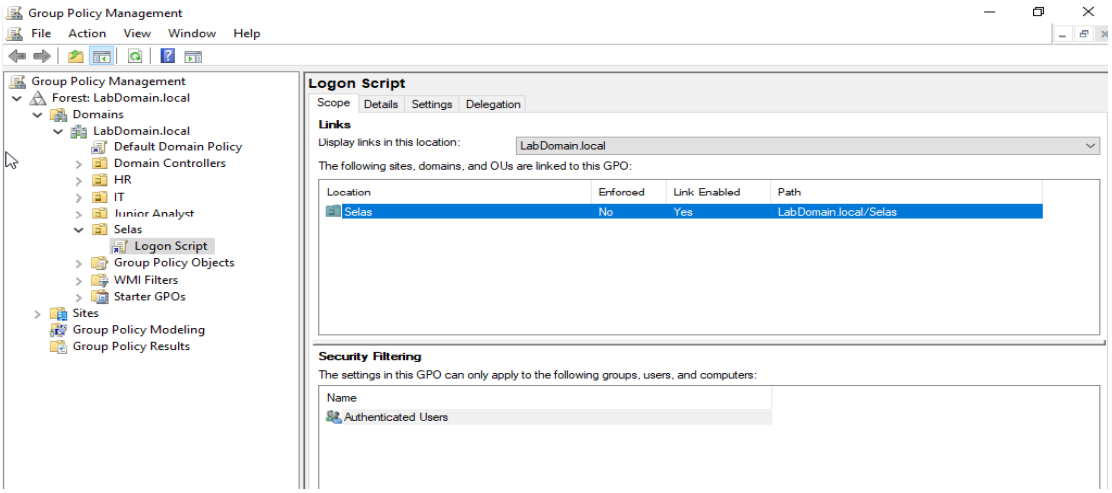
Password Policy - Before setting changes



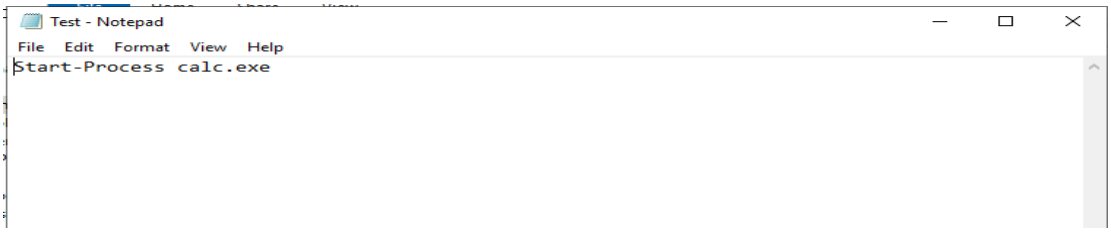
Password Policy - After setting changes



Logon Script GPO settings.



The Script



Exercise 5: Implement Advanced Group Policies

Create an AppLocker Policy GPO:

- In GPMC, create a new GPO named AppLocker Policy.
- Edit the GPO and configure AppLocker rules:
- Navigate to Computer Configuration -> Policies -> Windows Settings -> Security Settings -> Application Control Policies -> AppLocker.
- Create executable rules to allow only Microsoft-signed executables and block all others.

Create a Control Panel Restriction Policy GPO:

- In GPMC, create a new GPO named Control Panel Restriction.

Edit the GPO and configure Control Panel restrictions:

- Navigate to User Configuration -> Policies -> Administrative Templates -> Control Panel.
- Set Prohibit access to Control Panel and PC settings to Enabled.

Link the GPOs:

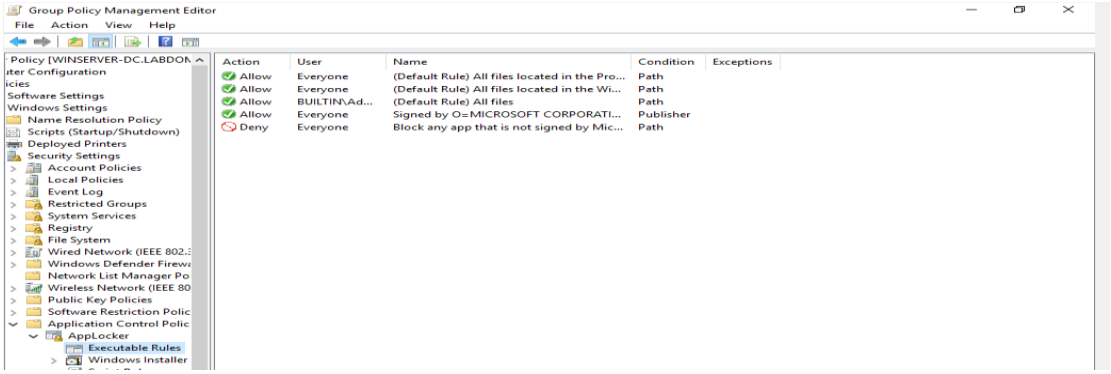
- Link the AppLocker Policy GPO to the domain LabDomain.local.
- Link the Control Panel Restriction GPO to the Sales OU.

Evidence Required:

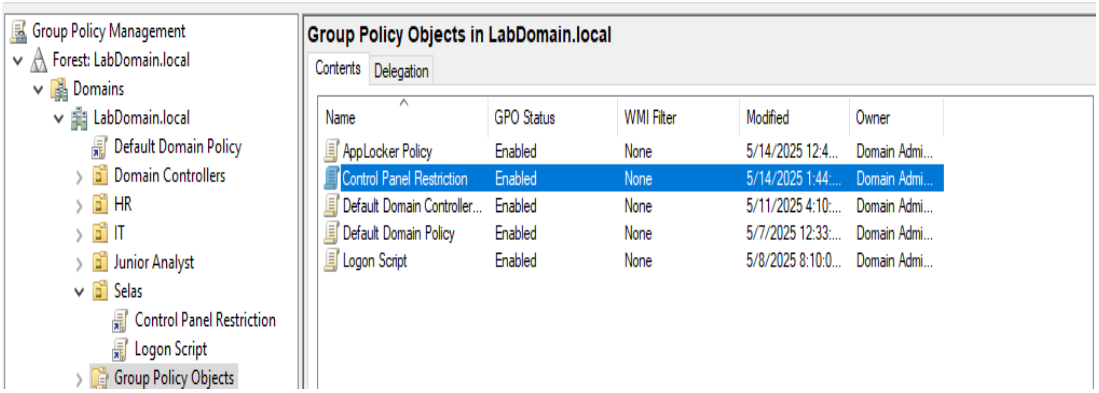
- Screenshot of GPMC showing the AppLocker Policy settings.
- Screenshot of Control Panel Restriction GPO settings.

Exercise 5: Implement Advanced Group Policies

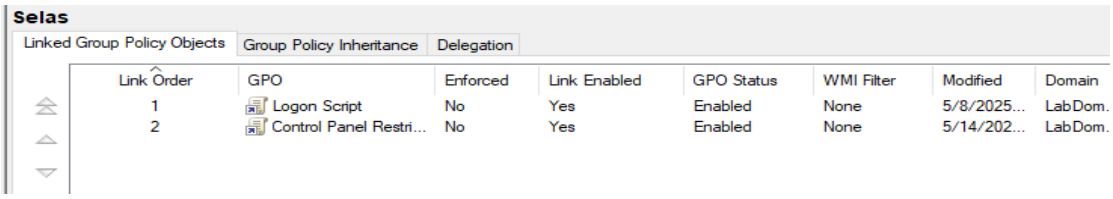
AppLocker Policy Settings



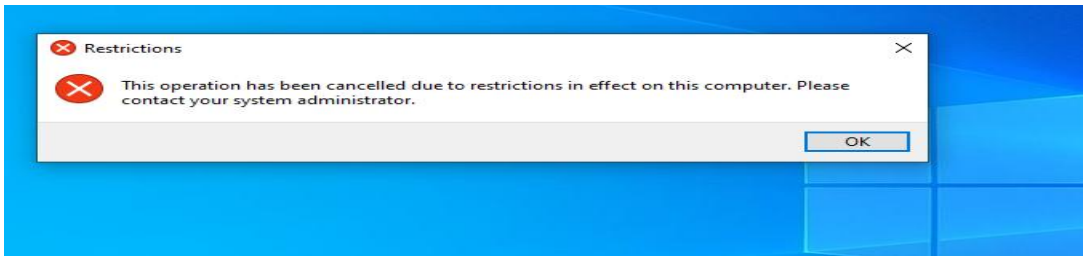
Control Panel Restriction GPO settings



Control Panel Restriction GPO settings on the Seals OU



Blocking on Seals user on windows client



DON'T FORGET TO MAKE A SNAPSHOT BEFORE THIS STEP !!!!!!!

Exercise 6: Backup and Restore Active Directory

Create a System State Backup:

- Install 'Windows Server Backup' if not already installed.
- Open 'Windows Server Backup'.
- Click 'Backup Once'.
- Choose 'Custom' and select 'System State'.
- Choose a backup destination and start the backup process.

Restore Active Directory:

- Boot into Directory Services Restore Mode (DSRM).
- Open 'Windows Server Backup'.
- Click 'Recover', choose the backup location and the appropriate backup.
- Select 'System State' and restore.

Evidence Required:

- Screenshot of 'Windows Server Backup' showing the completed backup.
- Screenshot of 'Windows Server Backup' showing the restore process.

Exercise 6 - Backup and Restore Active Directory

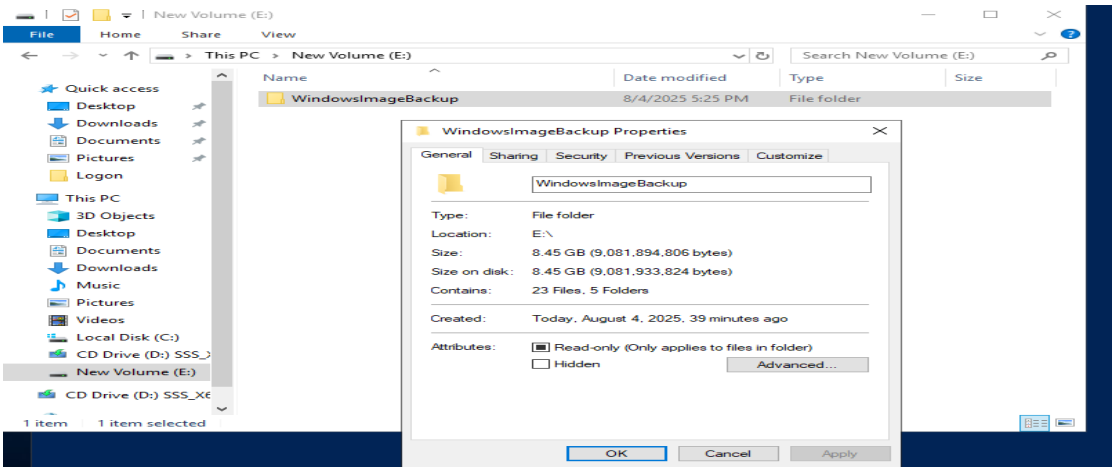
The Beginning of the backup process

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> wbadmin start systemstatebackup -backupTarget:E: -quiet
wbadmin 1.0 - Backup command-line tool
(C) Copyright Microsoft Corporation. All rights reserved.

Starting to back up the system state [8/4/2025 5:25 PM]...
Retrieving volume information...
This will back up the system state from volume(s) (C:) to E:.
Creating a shadow copy of the volumes specified for backup...
Creating a shadow copy of the volumes specified for backup...
Please wait while system state files to back up are identified.
This might take several minutes...
Found (241) files.
Found (3484) files.
Found (9608) files.
Found (17532) files.
Found (31955) files.
Found (37833) files.
Found (39791) files.
Found (41993) files.
Found (43973) files.
Found (46977) files.
Found (53137) files.
```

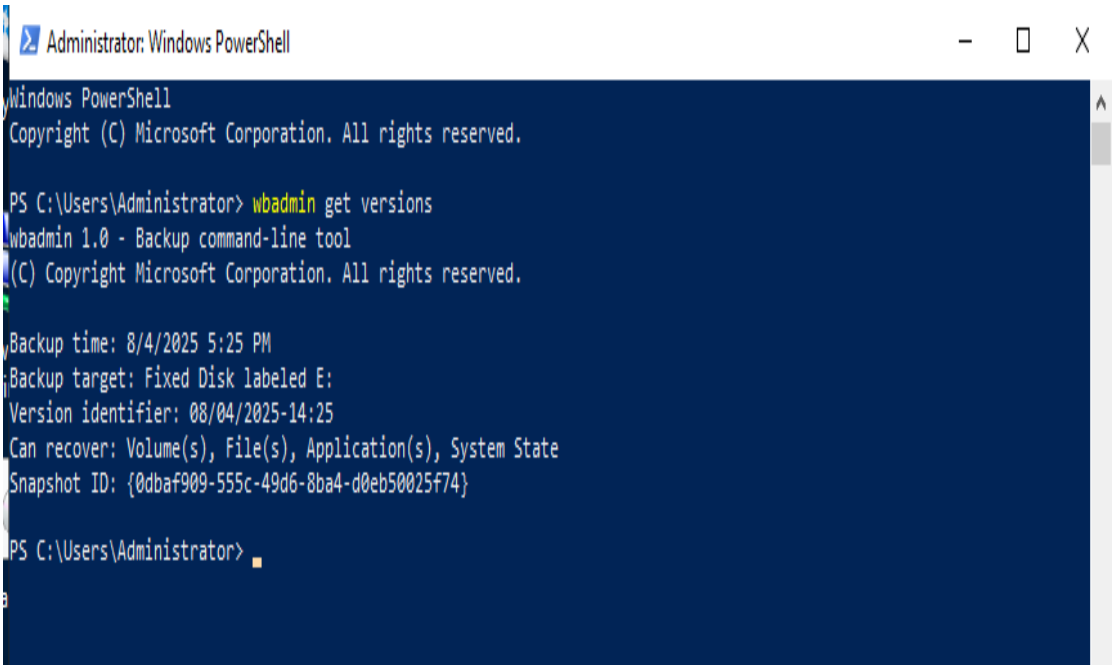
Backup process completed

```
Currently backing up files reported by 'System Writer'...
Overall progress: 96%.
Currently backing up files reported by 'System Writer'...
Summary of the backup operation:
------------------

The backup operation successfully completed.
The backup of the system state successfully completed [8/4/2025 2:59 PM].
Log of files successfully backed up:
C:\Windows\Logs\WindowsServerBackup\Backup-04-08-2025_14-25-35.log

PS C:\Users\Administrator>
```

The Path of the backup



The backup was successful. The restore was not performed due to DSRM connection limitations, but it was verified that the backup exists and is available for restore via wbadmin.



Exercise 7: Install and Configure Certificate Services

Install AD CS:

- Open Server Manager and click 'Add Roles and Features'.
- Select 'Active Directory Certificate Services'.

Choose the following services:

- Certification Authority
- Certification Authority Web Enrollment

Configure Certificate Services:

- Click 'Configure Active Directory Certificate Services'.
- Select 'Certification Authority' and 'Certification Authority Web Enrollment'.
- Choose 'Enterprise CA' and 'Root CA'.
- Create a new private key with a 2048-bit key length.
- Set the validity period for the CA certificate to 10 years.
- Specify database locations and complete the setup.

Verify the CA Installation:

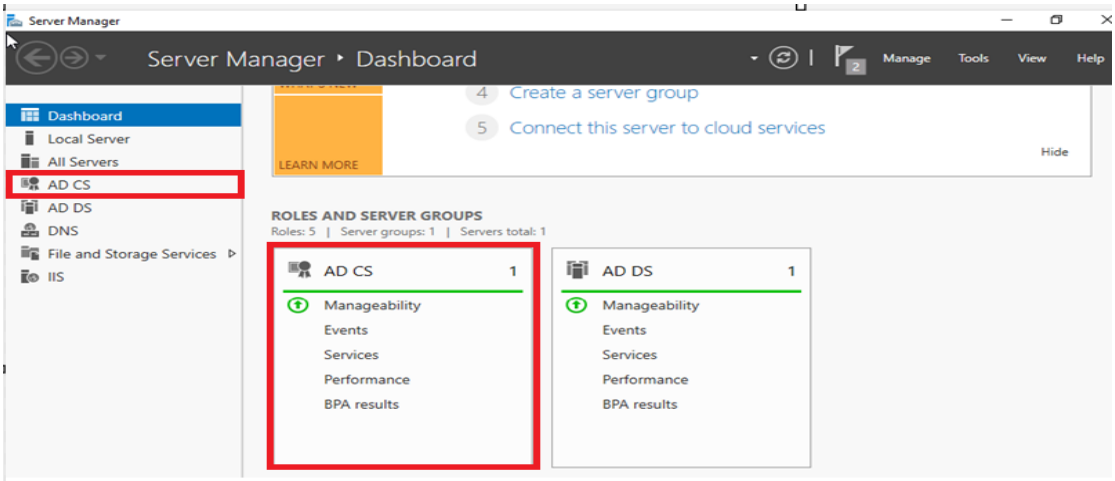
- Open 'Certification Authority'.
- Confirm the CA service is running.

Evidence Required:

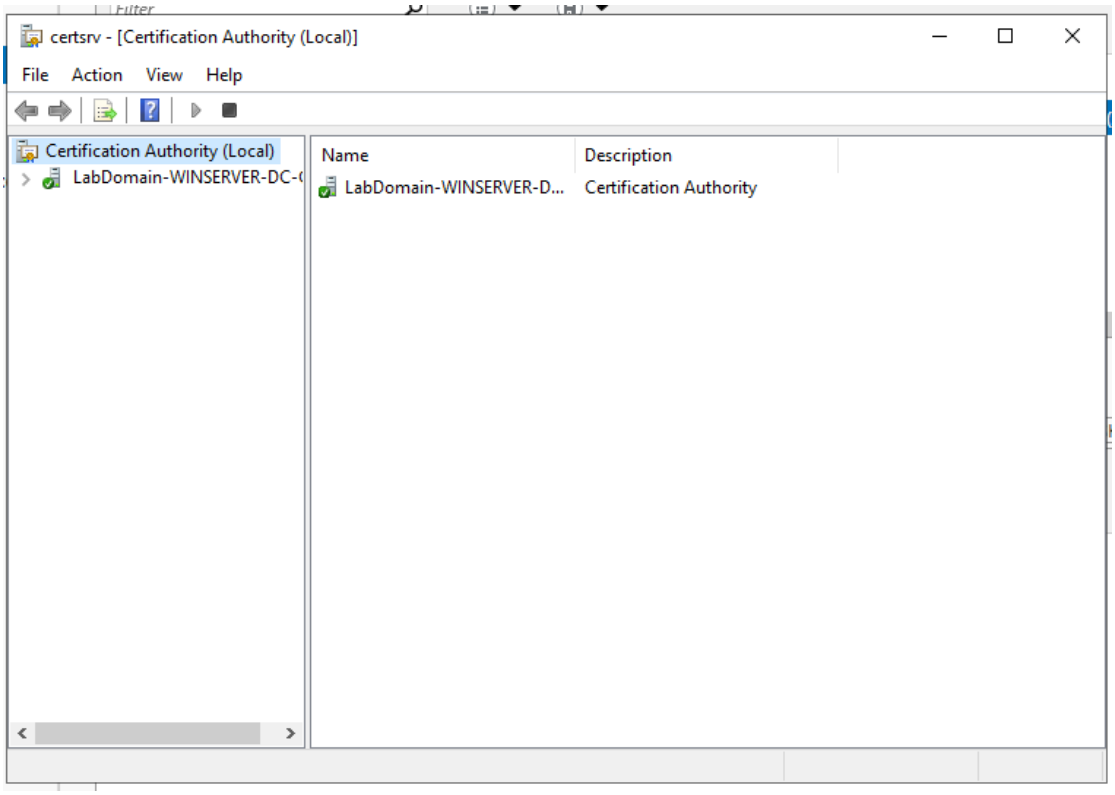
- Screenshot of Server Manager showing AD CS roles installed.
- Screenshot of 'Certification Authority' console showing the running CA.

Exercise 7: Install and Configure Certificate Services

Server Manager showing AD CS roles installed



Certification Authority in the console showing and running CA



Exercise 8: Create and Issue Certificates

Create a New Certificate Template:

- Open 'Certification Authority'.
- Right-click 'Certificate Templates' and select 'Manage'.
- Create a duplicate of the 'User' template.
- Name the new template CustomUserTemplate.

Configure the following settings:

- Validity period: 1 year
- Security permissions: Allow 'Domain Users' to enroll.

Issue the Certificate Template:

- Right-click 'Certificate Templates' in the 'Certification Authority' console.
- Select 'New' -> 'Certificate Template to Issue'.
- Choose CustomUserTemplate.

Enroll for Certificates:

- On WinClient-01, open 'certmgr.msc'.
- Right-click 'Personal' -> 'All Tasks' -> 'Request New Certificate'.
- Select the CustomUserTemplate certificate and enroll.

Evidence Required:

- Screenshot of 'Certification Authority' showing the issued template.
- Screenshot of 'certmgr.msc' on WinClient-01 showing the enrolled certificate.

Exercise 8: Create and Issue Certificates

During the configuration and testing of the Certificate Enrollment process, all the necessary steps were completed, But do to the lab limitations in OraclVitruslBox I couldn't complete the task.

All technical steps were correctly implemented (as you can see down below), and the issue is not caused by configuration errors but rather by limitations of the virtual lab environment. The failure to enroll the certificate is expected under these conditions.

RPC Endpoint Mapper (Port 135) & Dynamic RPC Ports:

Server

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> Get-Service RpcSs,CertSvc

Status   Name               DisplayName

Running  CertSvc            Active Directory Certificate Services
Running  RpcSs              Remote Procedure Call (RPC)


PS C:\Users\Administrator> netstat -ano | findstr 135
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING       112
  TCP    [::]:135               [::]:0                 LISTENING       112
  TCP    [fe80::2409:43fc:fa86:1f11%4]:49680  [fe80::2409:43fc:fa86:1f11%4]:135  TIME_WAIT       0
  TCP    [fe80::2409:43fc:fa86:1f11%4]:49683  [fe80::2409:43fc:fa86:1f11%4]:135  TIME_WAIT       0
  TCP    [fe80::2409:43fc:fa86:1f11%4]:49691  [fe80::2409:43fc:fa86:1f11%4]:135  TIME_WAIT       0
  TCP    [fe80::2409:43fc:fa86:1f11%4]:49701  [fe80::2409:43fc:fa86:1f11%4]:135  TIME_WAIT       0
  UDP    0.0.0.0:59135          *:*                                    2520
  UDP    0.0.0.0:60135          *:*                                    2520
  UDP    [::]:61135             *:*                                    2520
  UDP    [::]:61350             *:*                                    2520
  UDP    [::]:61351             *:*                                    2520
  UDP    [::]:61352             *:*                                    2520
  UDP    [::]:61353             *:*                                    2520
  UDP    [::]:61354             *:*                                    2520
  UDP    [::]:61355             *:*                                    2520
  UDP    [::]:61356             *:*                                    2520
  UDP    [::]:61357             *:*                                    2520
  UDP    [::]:61358             *:*                                    2520
  UDP    [::]:61359             *:*                                    2520
  UDP    [::]:62135             *:*                                    2520
  UDP    [::]:63135             *:*                                    2520
PS C:\Users\Administrator>
```

```
Administrator: Windows PowerShell
  UDP    [::]:63135             *:*                                    2520
PS C:\Users\Administrator> llo
llo : The term 'llo' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the
spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ llo
+ ~~~
    + CategoryInfo          : ObjectNotFound: (llo:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Administrator>
PS C:\Users\Administrator> net advfirewall firewall add rule name="Allow RPC Endpoint Mapper" dir=in action=allow protocol=TCP localport=135
The syntax of this command is:

NET
    [ ACCOUNTS | COMPUTER | CONFIG | CONTINUE | FILE | GROUP | HELP |
      HELPMSG | LOCALGROUP | PAUSE | SESSION | SHARE | START |
      STATISTICS | STOP | TIME | USE | USER | VIEW ]
PS C:\Users\Administrator> netsh advfirewall firewall show rule name="Allow RPC Endpoint Mapper"

Rule Name:                            Allow RPC Endpoint Mapper
----------------------------------------------------------------------
Enabled:                              Yes
Direction:                            In
Profiles:                             Domain,Private,Public
Grouping:
LocalIP:                              Any
RemoteIP:                             Any
Protocol:                             TCP
LocalPort:                            135
RemotePort:                           Any
Edge traversal:                       No
Action:                               Allow

Rule Name:                            Allow RPC Endpoint Mapper
----------------------------------------------------------------------
Enabled:                              Yes
Direction:                            In
Profiles:                             Domain,Private,Public
Grouping:
LocalIP:                              Any
RemoteIP:                             Any
Protocol:                             TCP
LocalPort:                            135
RemotePort:                           Any
Edge traversal:                       No
Action:                               Allow
Ok.

PS C:\Users\Administrator>
```

Client

```
PS C:\Users\Administrator> Test-NetConnection -ComputerName WinServer-DC.LabDomain.local -Port 135


ComputerName     : WinServer-DC.LabDomain.local
RemoteAddress    : 192.168.0.100
RemotePort       : 135
InterfaceAlias   : Ethernet
SourceAddress    : 192.168.0.20
TcpTestSucceeded : True



PS C:\Users\Administrator>
PS C:\Users\Administrator> Test-NetConnection -ComputerName WinServer-DC.LabDomain.local -Port 49158
WARNING: TCP connect to (192.168.0.100 : 49158) failed


ComputerName           : WinServer-DC.LabDomain.local
RemoteAddress          : 192.168.0.100
RemotePort             : 49158
InterfaceAlias         : Ethernet
SourceAddress          : 192.168.0.20
PingSucceeded          : True
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded       : False
```

Using Test-NetConnection we confirmed that TCP port 135 on the Domain Controller is open (TcpTestSucceeded : True).
However, testing a dynamic RPC port (49158) failed (TcpTestSucceeded : False), showing that dynamic ports were blocked in the environment.

Firewall configuration for RPC (Port 135):

Server

```
PS C:\Users\Administrator> netsh advfirewall firewall show rule name="Allow RPC Endpoint Mapper"

Rule Name:                            Allow RPC Endpoint Mapper
----------------------------------------------------------------------
Enabled:                              Yes
Direction:                            In
Profiles:                             Domain,Private,Public
Grouping:
LocalIP:                              Any
RemoteIP:                             Any
Protocol:                             TCP
LocalPort:                            135
RemotePort:                           Any
Edge traversal:                       No
Action:                               Allow

Rule Name:                            Allow RPC Endpoint Mapper
----------------------------------------------------------------------
Enabled:                              Yes
Direction:                            In
Profiles:                             Domain,Private,Public
Grouping:
LocalIP:                              Any
RemoteIP:                             Any
Protocol:                             TCP
LocalPort:                            135
RemotePort:                           Any
Edge traversal:                       No
Action:                               Allow

Rule Name:                            Allow RPC Endpoint Mapper
----------------------------------------------------------------------
Enabled:                              Yes
Direction:                            In
Profiles:                             Domain,Private,Public
Grouping:
LocalIP:                              Any
RemoteIP:                             Any
Protocol:                             TCP
LocalPort:                            135
RemotePort:                           Any
Edge traversal:                       No
Action:                               Allow
Ok.
```

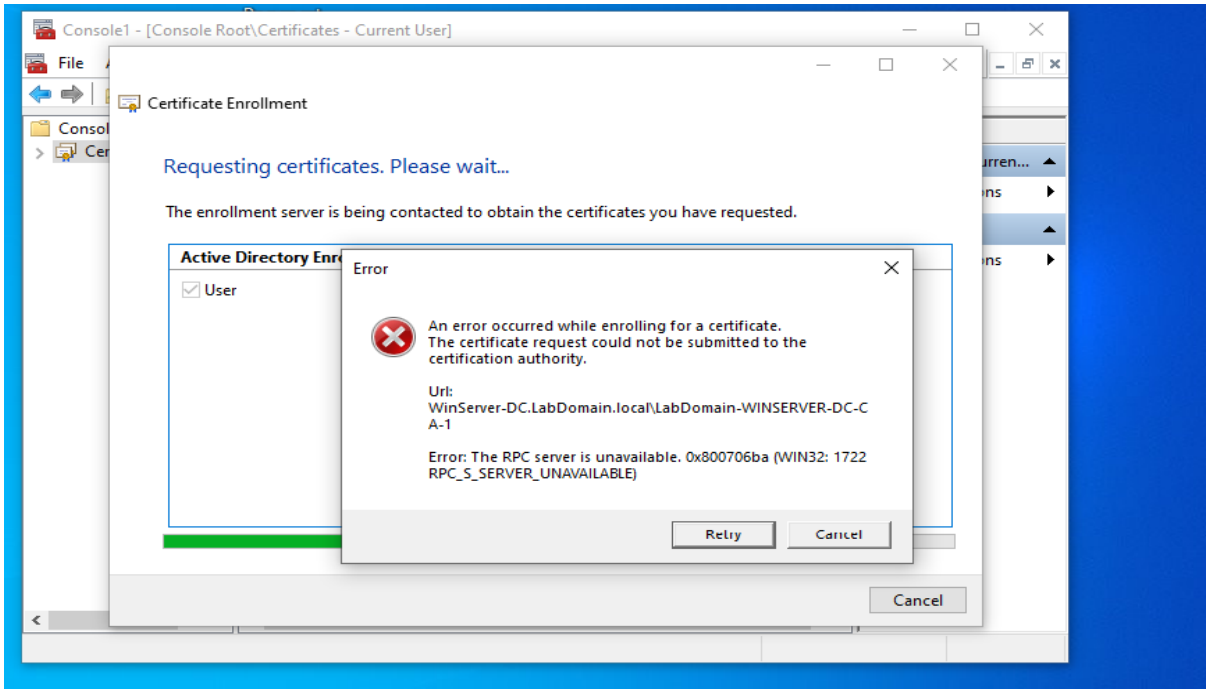
```
PS C:\Users\Administrator> netstat -ano | findstr 135
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING       112
  TCP    [::]:135               [::]:0                 LISTENING       112
  UDP    0.0.0.0:59135          *:*                                    2520
  UDP    0.0.0.0:60135          *:*                                    2520
  UDP    [::]:61135             *:*                                    2520
  UDP    [::]:61350             *:*                                    2520
  UDP    [::]:61351             *:*                                    2520
  UDP    [::]:61352             *:*                                    2520
  UDP    [::]:61353             *:*                                    2520
  UDP    [::]:61354             *:*                                    2520
  UDP    [::]:61355             *:*                                    2520
  UDP    [::]:61356             *:*                                    2520
  UDP    [::]:61357             *:*                                    2520
  UDP    [::]:61358             *:*                                    2520
  UDP    [::]:61359             *:*                                    2520
  UDP    [::]:62135             *:*                                    2520
  UDP    [::]:63135             *:*                                    2520
PS C:\Users\Administrator>
```

Client

```
PS C:\Users\Administrator> Test-NetConnection -ComputerName WinServer-DC.LabDomain.local -Port 135


ComputerName     : WinServer-DC.LabDomain.local
RemoteAddress    : 192.168.0.100
RemotePort       : 135
InterfaceAlias   : Ethernet
SourceAddress    : 192.168.0.20
TcpTestSucceeded : True



PS C:\Users\Administrator>
```

I enabled RPC communication by creating inbound firewall rules for TCP 135 (RPC Endpoint Mapper) and TCP 49152–65535 (dynamic RPC ports). After verifying that AD CS and RPC services were running, we tested from the client and confirmed that port 135 was open, while dynamic ports were blocked.

Certificate Enrollment Error Documentation



During certificate enrollment the client (User) I received constantly error massage 0x800706ba (RPC_S_SERVER_UNAVAILABLE), This shows that the DC was reachable but RPC communication failed,
most likely due to blocked dynamic RPC Ports or Firewall rules.